

Strength Training in Climbing: A Systematic Review

Kaja Langer, Christian Simon, and Josef Wiemeyer

Department of Human Sciences, Institute of Sports Science, Technical University Darmstadt, Darmstadt, Germany

Abstract

Langer, K, Simon, C, and Wiemeyer, J. Strength training in climbing: A systematic review. *J Strength Cond Res* 37(3): 751–767, 2023—The aim of this review was to provide an overview of the state of research on strength training in climbing and to answer the question how climbing performance, maximum grip strength, upper-limb strength endurance, maximum upper-limb strength, and upper-limb power as dependent variables are affected by different types of training. Moreover, we addressed the question which training methods and training parameters are most effective in increasing climbing and bouldering performance. Searches of MEDLINE (PubMed), SPORTDiscus, ProQuest, and Google Scholar were conducted for studies that met the following criteria: (a) examining effects of training on at least one of the dependent variables, (b) controlled longitudinal design with pretest and posttest, and (c) detailed information on training parameters and subjects. Twelve studies were included into the review. The quality of the studies was rated according to the PEDro scale, and the training interventions were classified according to training method (maximum strength [MS], hypertrophy [HYP], and endurance [END]), specificity (specific, semispecific, and unspecific), and static or dynamic exercises. For 9 of the 12 studies, effect sizes were calculated and the treatments compared. The results showed (a) positive effects of strength training on all variables, (b) a trend toward a mixture of MS and HYP or END training, (c) a trend toward semispecific exercise, and (d) similar effects for dynamic and static exercise with a trend toward a mixture of both. Coaches and athletes are recommended to combine static and dynamic semispecific exercises in a HYP and MS or END training.

Key Words: performance, specificity, training methods, strength endurance, power, grip strength

Introduction

Recently, climbing and bouldering have been included in the Olympic canon (15). The accompanying professionalization and the complex demands of the sport led to an increasing number of coaches and athletes looking for the evidence-based training guidelines. Several studies addressing the potential determinants of climbing have found that climbing performance is influenced by several trainable variables, such as grip strength-to-mass ratio, hip flexibility, upper-limb strength, finger and forearm strength, upper-limb rate of force development, and above-average endurance of the shoulder girdle (2,8,9,19,30,37,44). Among other factors, strength plays a substantial role in climbing. For example, elite climbers were found to have higher grip strength compared with skilled climbers and novices (6,19,40). In addition, maximum grip strength and upper-limb power were found to be higher in elite boulderers compared with elite climbers (6,26). According to Saul et al. (37), especially the strength-to-mass ratio and the strength endurance ability of the forearm, flexors play a prominent role in climbing success.

Because a substantial percentage of the variance of climbing and bouldering performance can be explained by trainable variables (46% (19) and 58.9% (30)), researchers (30,37) emphasized the importance of engaging in a training program to improve muscular strength, power, and strength endurance of the upper limbs. In particular, the demand for training recommendations to maximize performance has been increasing (30).

Yet, until today, most training recommendations are primarily based on traditional training methods and practical experience (9,45,47). Therefore, several authors complained about a lack of research in the field of training in climbing (33,47). In addition, current training recommendations (13,31) differ for recommended training methods and devices.

Considering the research deficits mentioned above, the aim of this review was to provide an overview of the current state of research on strength training in climbing and bouldering. In particular, the question is addressed how climbing performance, maximum grip strength (maximum strength of the finger flexors), upper-limb strength endurance, maximum upper-limb strength (maximum strength of the arms and shoulder girdle), and upper-limb power are influenced by different types of training. Furthermore, the review addresses the question, which training methods and training/exercise parameters are most effective to increase the performance of climbers and boulderers of different levels of competence. Finally, it is our goal to give coaches and athletes evidence-based training advice and to formulate recommendations for future research.

Methods

Search Strategy

We performed a systematic review of studies investigating the effects of strength training (independent variable) on climbing or bouldering performance and strength indicators, i.e., maximum grip strength, upper-limb strength endurance, maximum upper-limb strength, and upper-limb power (dependent variables).

The literature search was performed in September 2021, using the databases MEDLINE (PubMed), SPORTDiscus, ProQuest, and Google Scholar. The reference lists of articles retrieved in the

Address correspondence to Kaja Langer, kaja.langer@gmx.de.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (<http://journals.lww.com/nsca-jscr>).

Journal of Strength and Conditioning Research 37(3):751–767

© 2022 National Strength and Conditioning Association

search were subsequently manually searched for any additional articles that had potential applicability to the topic. An additional search in the resource pool of the international rock climbing research association (IRCRA) was performed. The search comprised the following terms: “climb,” “boulder,” “performance,” “strength,” “endurance,” “power,” “resistance,” “grip strength,” “hand strength,” “grip force,” and “finger flexors.” To maximize and optimize the detection of relevant studies, the terms were combined with the wildcard symbol “*” and linked by Boolean operators (OR and AND).

Study Inclusion and Exclusion Criteria

In this review, only peer-reviewed studies, published abstracts of a presentation at a scientific conference, and dissertations with a controlled longitudinal design and including at least 1 pretest and posttest were considered. Studies were included if they examined the effects of a maximum strength, hypertrophy, power, or strength endurance training in climbing or bouldering on at least one of the following dependent variables: (a) climbing or bouldering performance, (b) maximum grip strength, (c) upper-limb strength endurance of the upper limbs, (d) upper-limb power, and (e) maximum upper-limb strength and compared them with the effects of a no-treatment control group or with the effects of at least another training method.

In addition, studies had to fulfill the following inclusion criteria: (a) representative samples with human climbers/boulderers; (b) detailed information on training methods and parameters; (c) detailed information regarding the subjects (e.g., age, body mass, and climbing level); and (d) reported climbing levels, using a recognized national or international scale.

Cohort studies, case studies, cross-sectional studies, anecdotal research, and qualitative research do not allow causal interpretations because of their lower methodical quality and were therefore not included in the review. Because we were interested in the effects of strength training on human climbing performance, animal-based research was also excluded.

Risk of Bias Assessment

To ensure a reliable assessment of the quality of the studies, each study selected in the previous phase was fully assessed independently by the 3 reviewers using an adapted PEDro scale (27). Criteria 5–7 of the scale were excluded because they relate to blinding of subjects and investigators, which is not possible in a training intervention. After assessing all the studies, we also excluded criterion 3 from the assessment because it was not met by any of the studies. The maximum score in the risk of bias assessment was thus 7. Disagreements were resolved by consensus. To ensure interrater reliability, all 3 authors evaluated one of the studies selected in advance. The results of the following discussion formed the basis for the evaluation of all studies selected.

Classification of Exercises and Training Methods

All exercises were rated according to their specificity to climbing and bouldering. The specificity was divided into unspecific, semispecific, and specific according to Brearley and Bishop (4). Table 1 presents which exercises were assigned to which category.

The training methods were classified as maximum strength (MS), hypertrophy (HYP), and strength endurance training (END) according to the number of repetitions per set (dynamic exercises [dyn]) or the duration of 1 repetition (static exercises

[stat]) in line with the findings of multiple studies (21,25,35,38) (Table 2).

Data Extraction and Analysis

A standardized data extraction form was adopted, and the main characteristics of the selected studies were described in detail. The extraction form comprised study design, sample size, sample characteristics, and variables related to the interventions and the outcomes of the effects of interest. To allow comparison of the results between studies, reported grades for climbing and bouldering performance were transformed according to the International Rock Climbing Research Association (IRCRA) reporting scale (5). To avoid the inclusion of different publications with identical study groups, the time and place of recruitment and the composition of the test groups were examined, and the authors contacted for clarification if necessary.

The effects of different training interventions were statistically compared regarding specificity of exercises, training methods, and subjects' performance level. For studies comparing a treatment to a nontreatment group (11,12,16,17,20,32,34,40), outcome values referring to maximum grip strength, climbing performance, upper-limb power and strength, and upper-limb strength endurance of the upper limbs were extracted to calculate effect sizes. The same was performed for 2 studies comparing one training method with another because one intervention was similar to conventional climbing or bouldering and these groups, thus, were as controls (28,29). Data presented only graphically were extracted numerically with the help of a superimposed grid.

The analysis of the effects included meta-analytical elements to visualize the results. However, we did not conduct a meta-analysis because the number of studies did not meet the general recommendations for a statistically powerful meta-analysis (3,10,42). Effect sizes (d_{Cohen} : 0.2—small, 0.5—moderate, and 0.8—large) and confidence intervals (95%) were calculated according to Fritz et al. (7). No effect sizes were calculated for studies comparing 2 different treatments (22,24,34). To be able to compare the results of the different studies regarding the 5 dependent variables selected, separate mean values of the calculated effect sizes and confidence intervals were calculated for each variable. For each study, the mean values represented at least one of a total of 5 groups: (a) upper-limb strength endurance (dead hang [s], bent arm hang [s], intermittent finger hangs [s], and climbing time to exhaustion [s]), (b) climbing performance (elevation points, American Sport Climbing Federation [ASCF] points, and number of completed moves) in both climbing and bouldering, (c) upper-limb power (absolute rate of force development ($\text{RFD}_{95\%}$) [$\text{N}\cdot\text{s}^{-1}$]: measured between the onset and the last slope of the isometric contraction curve) and explosive muscle force ($\text{RFD}_{200\text{ms}}$) [$\text{N}\cdot\text{s}^{-1}$]: measured within the first 200 ms of contraction), (d) maximum grip strength (hand dynamometry [kg]), and (e) maximum upper-limb strength (isometric pull-up force [kg]). The results were then compared with each other according to the specificity of the exercises or the method chosen in the treatment and visualized by means of forest plots.

Results

Study Selection

The search of the databases provided a total of 937 references. In addition, 17 studies were identified through manual search in the reference lists of articles retrieved in the search and through hand

Table 1
Classification scheme for training exercises.

	Unspecific	Semispecific	Specific
Criteria	General preparatory exercise Task to develop capacities, only low kinematic and dynamic similarity to the target task	Specific preparatory exercise High kinematic and dynamic similarity to the target task	Target task Identical with target task
Exercises	Pull-down Seated bench press Seated rowing Seated shoulder press Biceps curl Forearm press Forearm curl	Dead hangs and bent arm hangs from different holds/a fingerboard (with added weight) Fingerboard training of any kind Campus board training Different kinds of pull-ups	Interval bouldering/climbing Lead climbing Bouldering

search in the resource pool of the IRCRA. After duplicates were removed, 720 citations remained. Of these, 695 were excluded for the abovementioned reasons after reviewing the titles and abstracts. After a full-text review of the remaining 25 studies, 12 studies meeting the inclusion criteria were included in the analysis (Figure 1).

Description of the Studies

Table 3 summarizes the study design, subject characteristics, interventions, outcome values, the PEDro score, and the classification of the training methods and the treatments of the studies 12 ultimately included in this review comprising a total of 273 subjects. A less complex visualization is provided in the supplementary material (see Table s7, Supplemental Digital Content 1, <http://links.lww.com/JSCR/A334>).

Two studies focused on beginners (16,17), 3 on recreational climbers (11,12,32), 2 on advanced competitive and non-competitive climbers (34,40), and 5 on elite athletes (20,22,24,28,29). The percentage distribution of climbing levels among all subjects is illustrated in Figure 2. One study included only women (17), 5 studies only men (16,20,28,29,40), and 6 included a mixed sample (11,12,22,24,32,34). Six studies reported that none of the subjects had been engaged in a periodized training at least 1 month before this study (22,24,28,29,32,40). All subjects were aged between 16 and 40 years, had given written informed consent, and were healthy and without injury.

Risk of Bias

The risk of bias assessment resulted in a PEDro score of 7/7 for 5 studies (28,29,32,34,40) and a score of 6/7 for 5 studies (12,16,20,22,24). The quality of these studies was thus estimated as good. Two of the studies (22,24) with a PEDro score of 6/7 did not meet criterion 8 of the PEDro scale because no more than 85% of the subjects originally assigned to the groups had at least 1 key outcome measured. Of the other studies rated 6/7, one (12) did not meet criterion 1 because no

eligibility criteria were specified, one (16) did not meet criterion 2 because the subjects were not randomly allocated to the groups, and one (20) did not meet criterion 4 because the groups were not similar about the key prognostic indicators. The quality of 2 studies (11,17) was rated as moderate. One study (17) received a PEDro score of 5/7 because no eligibility criteria were specified (criterion 1), and the subjects were not randomly allocated to the groups (criterion 2). The other study (11) was rated 4/7 because it did not meet criteria 4, 8, and 9. The groups were not similar about the key prognostic indicators, at least 1 key outcome was not measured by more than 85% of the subjects originally assigned to the groups, and not all subjects for whom outcome measures were available received the treatment or control condition as allocated or, where this was not the case, data for no key outcome were analyzed by “intention to treat.” The detailed PEDro scores are presented in Table s6, Supplemental Digital Content 1 (<http://links.lww.com/JSCR/A334>).

Classification of Exercises and Training Methods

Of the 12 studies selected, 5 compared the effects of 1 treatment and 4 compared the effects of 2 different treatments with a no-treatment control group. One study compared 3 different treatments, and 2 studies compared 2 different treatments. Of the treatments performed in the studies, 4 (19.05%) were classified as unspecific, 3 as specific (14.29%), 13 as semispecific (61.9%), and 1 as a mixture of specific and semispecific treatments (4.76%).

Training methods were classified as dynamic endurance training (END) for 3 treatment groups (14.29%). One group (4.76%) combined END and hypertrophy training (HYP). A total of 9 treatments (42.86%) were identified as HYP, of which 1 included only dynamic and 8 included only static exercises. Three groups (14.29%) trained for maximum strength (MS) (2 dynamic and 1 static). The remaining 2 treatment groups both trained with a mixture of MS and HYP training, while one of the groups combined dynamic and static methods and the other trained only statically.

Table 2
Classification scheme for training methods.

Method	Number of repetitions per set (dynamic)	Duration of one repetition (static)
Strength endurance	>15	>30 seconds
Hypertrophy	8–15	3–30 seconds
Maximum strength	1–5	1–5 seconds

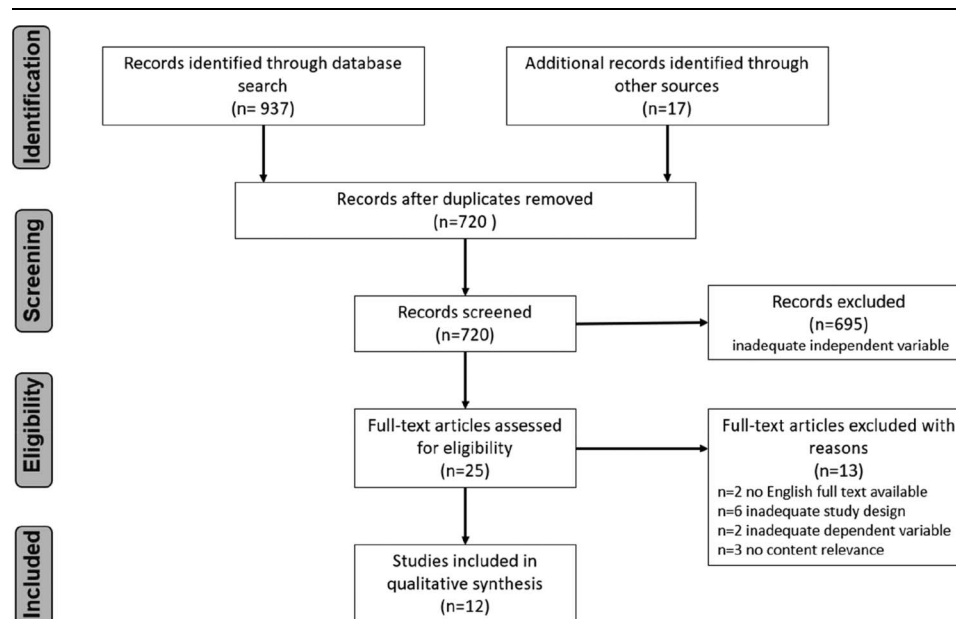


Figure 1. PRISMA flowchart of data extraction from the literature search.

Effects of Interventions

Six studies (11,12,16,17,34,40) assessed the effects of a treatment on climbing performance. Six studies (11,12,22,24,28,29) measured changes in the subjects' upper-limb strength endurance, 6 (16,17,20,22,28,32) examined differences in maximum grip strength, 2 (20,40) focused on the rate of force development in the upper limbs, and 1 (40) assessed maximum upper-limb strength.

Four studies performed an additional intervention check: Hermans et al. (11,12) checked the efficacy of their treatments by measuring the improvement in 1 exercise included in the training regimen, Stien et al. (40) measured campus board performance before and after 5 weeks of campus board training, and Philippe et al. (34) measured lactate concentrations and heart rate 1 and 6 minutes after climbing.

For a total of 9 studies (11,12,16,17,20,28,29,32,40), effect sizes were calculated. Four of these measured upper-limb strength endurance (11,12,28,29), 5 climbing performance (11,12,16,17,40), 5 maximum grip strength (16,17,20,28,32), 2 upper-limb power (20,40), and 1 maximum upper-limb strength (40). The results are presented as standardized mean differences (SMDs) between the treatment and control groups with 95% confidence intervals (CIs) in Figures 3–7. The exact values can be found in the supplemental material (Table s1–Table s5, Supplemental Digital Content 1, <http://links.lww.com/JSCR/A334>).

Climbing Performance

Of the 6 studies measuring climbing performance, 3 observed a significant improvement in the experimental groups (12,16,17). Effect sizes were calculated for 5 studies (11,12,16,17,40). The results are shown in Figure 3.

Kingsley (17) reports the only statistically significant effect size ($d = 1.0$). The treatment group, which consisted of beginners, had performed semispecific, static exercises consisting of a mixture of maximum strength and hypertrophy training methods for 7 weeks. However, these results should be taken with caution because the methodological quality of this study was found to be

very low. A similar training program performed by Jurens (16) resulted in a small effect size ($d = 0.44$). Other training programs that included unspecific and dynamic exercises and MS or END methods only (11,12) led to negligible to small effect sizes ($d = 0.12$, $d = 0.24$, $d = 0.31$, and $d = 0.38$, respectively). The methodological quality of the study by Hermans et al. (11) was not very high, and the results show no exceptionally high or low effect sizes. A mixture of MS and HYP methods with semispecific, dynamic exercises also led to small effect sizes ($d = 0.23$ and $d = 0.34$) (40).

According to the results of the calculated effect sizes, a mixture of MS and HYP training including static, semispecific exercises led to the strongest effect sizes in climbing performance while the same mixture of methods with dynamic, semispecific exercises led to slightly smaller effect sizes.

Philippe et al. (34) examined the effects of specific END and semispecific/specific HYP training on climbing performance in advanced climbers. Only nonsignificant improvements in the number of climbing moves, climbing difficulty, climbing time, and moves per time were found for both groups while a significant effect over time for climbing performance in both treatment groups was reported.

Upper-limb Strength Endurance

The strongest, significant effect size in upper-limb strength endurance was found by Medernach, Kleinöder, and Lötzerich (28) ($d = 1.6$) for a training group that trained both dynamically and statically and combined semispecific exercises with MS and HYP methods (Figure 4). Three further treatments led to significant positive large effect sizes. These treatments include dynamic, unspecific exercises according to a pure MS or END training (11) ($d = 1.4$ and $d = 1.0$) and dynamic, specific exercises incorporated into a mixture of a HYP and END training (29) ($d = 1.0$). The effect sizes of the study conducted by Hermans et al. (11), however, need to be treated with caution because of the study's low methodological quality. The effect sizes found for unspecific, dynamic HYP and END training by Hermans,

Table 3**Effects of different types of strength training on climbing or bouldering performance and multiple climbing specific tests.†**

Study	PEDro score	Study design	Subject (N, level, sex, age, and experience)	Treatment	Outcomes: Percent change pretest to posttest			
(11)	4	Randomized controlled trial (10 wk) Within-group and between-group design 1. MS dyn; unspecific ($n = 10$) 2. END dyn; unspecific ($n = 10$) 3. no-treatment control group ($n = 10$) pretest and posttest	30 climbers (m/f) 8–13 IRCRA Age: 20–28 Climbing experience 3.2 ± 2.3 y	2 sessions per wk 7 exercises (pull-down, seated bench press, seated rowing, seated shoulder press, biceps curls, forearm press, and forearm curl) 1. MS dyn; unspecific: 4 sets, 5RM, 3 min rest 2. END dyn; unspecific: 2 sets 20RM, 2 min rest between sets 3. CG: no additional strength training All groups were asked to maintain preintervention climbing volume	Climbing performance Elevation points [#] Strength endurance Dead hang [s] Bent arm hang [s] Treatment check 12 RM pull-down [kg]	1. MS dyn, unspecific +11.3 +84* +58.0* +16.5*	2. END dyn, unspecific +12.0 +87.5* +52.9* +17.7*	3. CG +4.5 +14 +17.1* +2.7
(12)	6	Randomized controlled trial (10 wk) 1. MS dyn; unspecific ($n = 10$) 2. END dyn; unspecific ($n = 11$) 3. no-treatment control group ($n = 10$) pretest and posttest	31 recreational climbers (m/f) 5–12 IRCRA Age 23.6 ± 2.2 y Climbing experience 3.2 ± 2.3 y	2 sessions per wk 7 exercises (pull-down, seated bench press, seated rowing, seated shoulder press, biceps curl, forearm press, and forearm curl) 1. MS dyn; unspecific: 4 sets, 5 reps, 3 min rest between sets 2. END dyn; unspecific: 2 sets, 20 reps, 2 min rest between sets CG: no special treatment: Continued climbing/training as usual	Climbing performance Elevation points [#] Strength endurance Dead hang [s], Bent arm hang [s] Treatment check 12M pull-down [kg]	1. MS dyn, unspecific +10.9* +86.7* +46.5* +17.4*	2. END dyn, unspecific +14.2 +87.5* +37.7* +15.5*	3. CG +9.5 +12.0 +14.3* +2.0
(16)	6	Quasi experimental between group Design (8 wk) 1. MS/HYP stat; semispecific ($n = 15$) 2. control group ($n = 11$) pretest and posttest	29 students (m) Beginners Age 18–26 y no experience	1. MS/HYP stat; semispecific: 12 sessions in total, 6 sets of dead hangs on 6 different holds, 5s on each hold (wk 1, 7s wk 2–4, 10s wk 5–8), 5s rest in between reps, 1min rest in between sets (feet supported on small edges below the hangboard, no additional training allowed) 2. CG: no special treatment: Regular class activities (climbing)	Maximum grip strength [kg] Climbing performance ASCF points [#]	1. MS/HYP stat, semispecific +4.5* +60.3	2. CG –3.0 +41.8	
Significant main effect in climbing performance when both groups' data were combined								
(17)	5	Quasi experimental between group design (7wk) 1. MS/HYP stat; semispecific ($n = 9$) 2. control group ($n = 12$) pretest and posttest	21 students (f) Beginners Age 18–26 y no experience	7 wk, 2 sessions per wk 1. MS/HYP stat; semispecific: 6 sets of dead hangs on 6 different holds, 5s on each hold (wk 1, 7s wk 2–4, 10s wk 5–8), 5s rest in between reps, 1min rest in between sets, encouraged to hang with full weight but support for feet allowed	Maximum grip strength [kg] Climbing performance ASCF points [#]	1. MS/HYP stat, semispecific +8.3* +82.7*	2. CG +4.5 +26.6	

Table 3**Effects of different types of strength training on climbing or bouldering performance and multiple climbing specific tests.† (Continued)**

Study	PEDro score	Study design	Subject (N, level, sex, age, and experience)	Treatment	Outcomes: Percent change pretest to posttest
(20)	6	Randomized controlled trial (4 wk) 1. MS stat; semispecific ($n = 7$) 2. no-treatment control group ($n = 7$) pretest and posttest	14 boulderers (m) >28 IRCRA Age: 21–29 Training experience ≤5 y	2. CG: no special treatment: Normal class activities (climbing) 1. MS stat, semispecific: 3 training sessions per wk, 2 sets of 6 exercises (1. Slope-, 2. + 3 half-, 4. slope-, 5. + 6. half-crimp, holds individually chosen so athlete could hold on 4–6s with 120° elbow), 3 min rest between exercises, other 3 training sessions per wk involved regular climbing training 2. CG: regular climbing 6 sessions per wk, but no specific training program	Maximum grip strength F_{max} [N] Upper-limb power RFD _{200ms} [N/s], RFD _{95%} [N/s] 1. MS stat, semispecific +6.6 _a +41* _a +16.5 _a 2. CG −1.8 _a +0.5 _a −0.8 _a
(22)	6	Randomized controlled trial (9 wk (2 × 4 including one rest week)) Crossover design 1. HYP stat; semispecific ($n = 5$) 2. HYP stat; semispecific ($n = 4$) Two pretests 1 and 2 wk before the start of the study, intermediate test wk 5, posttests wk 10, 12, 14	9 climbers (8m, 1f) ≥23 IRCRA Age ≥25 y Climbing experience ≥5 y	2 sessions per wk, 3–5 sets of 10s dead hang at half crimp with 3s reserve, 3min rest between sets wk 1–4 group 1 (HYP stat; semispecific): Minimum edge depth, group 2 (HYP stat; semispecific): Maximum added weight (wk 1: 3 sets; wk 2: 4 sets; wk 3–4: 5 sets) wk 6–9 group 1: Maximum added weight, group 2: Minimum edge depth (wk 6: 3 sets; wk 7: 4 sets; wk 8–9: 5 sets) - Minimum edge depth: half crimp, edge used that allowed 10s hangtime with 3 s margin, no added weight - Maximum added weight: 18-mm edge, half crimp, maximum weight added to allow 10s hangtime with 3s margin Technical and physical training parallel to intervention for both groups: 6 sessions per wk, 2–4 h, artificial climbing wall, 3–10 sets, 3–19 moves, 70–100% of max. Sport level When resting from study individual training at the performance level (wk 10–14)	Maximum grip strength [kg] Strength endurance Dead hand [s] 1. HYP stat, semispecific Minimum edge depth +2.1 _a +16.9 _a 2. HYP stat, semispecific Weight added +1.4 _a +20.0 _a
(24)	6	Randomized controlled trial (9 wk (2 × 4 including one rest week)) ABCCBA assignment procedure	26 climbers (23m, 3f) 22/23 IRCRA Age older than 25 y Climbing experience ≥2 y	6 sessions per wk Treatment design 2 × 4 wk, 2 sessions per wk, standardized warmup	strength Endurance Dead hang [s] 1. HYP stat, Semispecific Weight added +6.6 2. HYP stat, Semispecific Minimum edge depth 3. HYP stat, s Semispecific Mixture +34.1*

Table 3**Effects of different types of strength training on climbing or bouldering performance and multiple climbing specific tests.† (Continued)**

Study	PEDro score	Study design	Subject (N, level, sex, age, and experience)	Treatment	Outcomes: Percent change pretest to posttest	
		1. HYP stat; semispecific ($n = 11$) 2. HYP stat; semispecific ($n = 8$) 3. HYP stat; semispecific ($n = 7$) Pretest, intermediate test wk 5, posttest		1. HYP stat; semispecific: Maximum load dead hangs: Load in terms of extra weight or edge depth that allowed 13s hangtime with 3s margin before failure, 3–5 sets break (wk 1,5: 3 sets, wk 2,6: 4 sets, wk 3,4,7,8: 5 sets) of 10s with 3 min rest in between 8 (wk 1,5: 3 sets, wk 2,6: 4 sets, wk 3,4,7,8: 5 sets) 2. HYP stat; semispecific: Dead hangs on minimal edge depth without extra weight: 3–5 sets break (wk 1,5: 3 sets, wk 2,6: 4 sets, wk 3,4,7,8: 5 sets), 5 reps of 10 s, 5 s in between reps, 3 min in between sets, edge depth allowing 30s hangtime 3. HYP stat; semispecific: Exercise of group 2 in wk 1–4 (wk 1: 3 sets, wk 2: 4 sets, wk 3,4: 5 sets) and exercise of group 1 in wk 5–8 (wk 5: 3 sets, wk 6: 4 sets, wk 7,8: 5 sets) All subjects maintained regular climbing training (6 d a week)	+45.2*	
(28)	7	Randomized controlled trial (4 wk) - ABBA assignment procedure 1. MS/HYP dyn + stat; semi specific ($n = 12$) 2. MS/HYP dyn; specific ($n = 11$) Classified as control group pretest and posttest	23 boulderers (m) >20 IRCRA Age 25.6 ± 4.4 y Climbing experience >3 y	3 sessions per wk 1. MS/HYP dyn + stat; semispecific: Complex training regimen consisting of different exercises on different days: 4 wk, 3 sessions per wk, session duration 150 min, 48h rest between sessions group 1: 25–35 boulders, 4–8 handholds each, 5 min rest after each boulder group 2: Complex training regimen consisting of different exercises on different days: <i>Monday:</i> - FB repeaters: 6 reps (bent arm hang), 5 sets, 5s between hangs - FB hangs: 5 reps (1rep = 8s straight arm hang, 8s top lock off hang, 8s 120° lock off, 8s 90° lock off) 3 sets, 10s between hangs) - Uneven lock-offs: 5 reps (10s one arm hang lock off each side), 3 sets, 10s between reps	Maximum grip strength [kg] Strength endurance Dead hang [s], Intermittent finger hangs [s]	1. MS/HYP dyn + stat, semispecific +5.0 +106.3* _a +33.8 2. MY/HYP dyn, specific (CG) +2.3* +56.9* _a +1.2*

Table 3**Effects of different types of strength training on climbing or bouldering performance and multiple climbing specific tests.† (Continued)**

Study	PEDro score	Study design	Subject (N, level, sex, age, and experience)	Treatment	Outcomes: Percent change pretest to posttest			
				<i>Wednesday:</i> - FB hangs: 10 reps (3–5s straight arm hang), 3 sets, 30s between hangs - Frenchies: Pull-up, 5s lock off at top, lower down, same at 90° and then at 120° and 2 normal pull-ups - FB hangs: 8 reps (20s straight arm hang + 2 pull-ups), 3 sets, 10s between hangs <i>Friday:</i> - FB repeaters: 8 reps (10s straight arm hang), 5 sets, 2s between hangs - Lock offs: 6 reps (1rep = 5s straight arm hang and lock off right hand and same with left hand), 5 sets, no rest between hangs - FB pyramid: 5 reps (1 rep = 4; 6; 8; 10 s at 90° + 2 pull-ups), 3 sets, 10s between hangs 2. MS/HYP dyn; specific: 25–35 boulders, 4–8 handholds each, 5min rest after each boulder				
(29)	7	Randomized controlled trial (4 wk) 1. HYP/END dyn; specific (<i>n</i> = 12) 2. MS/HYP dyn; specific (<i>n</i> = 12) Classified as control group pretest and posttest	24 boulderers (m) 20-22- IRCRA Age 25.2 ± 4.8 y bouldering experience ≥3 y	3 sessions per wk 1. HYP/END dyn; specific: Interval patterns of 4, 6, 8 and 10 climbing movements, each pattern climbed 4 times = 1 set, 3 reps, no rest between reps, 7 min rest between sets, intensity chosen to guarantee post effort individual exhaustion of forearm muscles 2. MS/HYP dyn; specific: 25–35 boulder problems per session at individual ability level, 4–8 handholds per boulder (less than 1 min), 5min rest between each set	Strength endurance Intermittent finger hangs [s], climbing time to exhaustion [s]	1. HYP/END dyn, specific +29.3* +38.8*	2. MS/HYP dyn, specific (CG) +5.2 +6.6	
(32)	7	Randomized controlled trial (8 wk) 1. HYP stat; semispecific (<i>n</i> = 9) 2. HYP stat; semispecific (<i>n</i> = 8) 3. control group (<i>n</i> = 10) Pretest and posttest	27 climbers (15m, 12f) IRCRA 10–17 Age 24 ± 3.5 y. no hangboard experience	8wk, 2–3 sessions per wk 1. HYP stat, semispecific: Smallest grip that could be held for 10 s was set as starting level; complex training regimen with increasing training load by reducing edge depth (3 sets per session, 3 to 5 reps. per set with a duration of 3–10 s per rep and 10–60 s rest in	maximum Grip strength [kg]	1. HYP stat, semispecific Minimum edge depth +5.0 _a	2. HYP stat, semispecific Weight added +11.9* _a	2. CG +1.1 _a

Table 3**Effects of different types of strength training on climbing or bouldering performance and multiple climbing specific tests.† (Continued)**

Study	PEDro score	Study design	Subject (N, level, sex, age, and experience)	Treatment	Outcomes: Percent change pretest to posttest		
				between the reps.) (supplemental material x) 2. HYP stat, semispecific: Smallest grip that could be held for 10 s was set as starting level; complex training regimen with increasing training load by adding weight (3 sets per session, 3 to 5 reps. per set with a duration of 3–10 s per rep and 10–60 s rest in between the reps.) (supplemental material x) 3. CG: regular climbing training 2–3 sessions per week, no training on hangboard or fingerboard allowed All groups maintained regular climbing training			
(34)	7	Randomized controlled trial (8 wk) 1. HYP dyn; semispecific + specific (<i>n</i> = 11) 2. END dyn; specific (<i>n</i> = 12) Pretest, posttests after intervention and after a 2 wk tapering period	23 climbers (15m, 8f) ≥16 IRCRA Age 16 – 40 y Competitive and noncompetitive climbers	5 sessions per wk 1. HYP dyn; semispecific + specific: 3 sessions per wk: bouldering (3 exercises (pinch/sloper/crimp) 8 sets of 10 reps, 2 min rest between sets, 12min rest between exercises, load so that only 10 moves possible) and campus board training (2 exercises (1: 1-4-7 exercise, 5 sets left hand, 5 sets right hand 2: 5 pull-ups, blocking at 120°, 90°, 60° for 7s); 2 sessions per wk: Lead project climbing (4 routes, 30 moves each, not more than 10-12moves in a row possible, 20min rest between routes 2. END dyn; specific: Sessions 1 + 3: hard lead projects (3–4 sets of 15–30 moves in a row, only max. 30 moves possible before exhaustion, 20min rest between sets); extensive intervals (in wk 1 + 2 (1 route climbed 10 times in a row, exhaustion after 9–10 attempts) or intensive intervals (in wk 3–8 (2 blocks of top roping, 8 sets of 25 moves in a route topable 5 times, break between blocks 49–50min)) in session 3	Climbing performance Climbing moves [#], Climbing difficulty [IRCRA], Climbing time [s], Moves per time [#s] Physiological parameters Lactate post 1 min [mmol/l], Lactate post 6 min [mmol/l], Heart rate post 1 min [bpm], Heart rate post 6 min [bpm]	1. HYP dyn, semispecific + specific +4.9 +2.8 –9.0 +14.7 –3.1 –7.1 +3.0 +2.4	2. END dyn, specific +4.1 +1.5 –8.3 +14.6 –1.7 +3.7 –0.8 +0.1

Table 3**Effects of different types of strength training on climbing or bouldering performance and multiple climbing specific tests.† (Continued)**

Study	PEDro score	Study design	Subject (N, level, sex, age, and experience)	Treatment	Outcomes: Percent change pretest to posttest			
(40)	7	Randomized controlled trial (5 wk) 1. MS/HYP dyn; semispecific (<i>n</i> = 5) 2. MS/HYP dyn; semispecific (<i>n</i> = 6) 3. control group (<i>n</i> = 5) Pretest and posttest	16 climbers (m) IRCRA ≥18 Age ≥18 Multiple years of experience	Sessions 2 + 4: Climbing easy lead projects (30–50 moves. 3–4 times, 2° min rest in between Session 5: Basic lead climbing (8 submaximal routes) 2 wk tapering period: both groups climbed at low intensity climbing at one pace	Significant main effects over time but not time × group interaction			
				5wk; 2 versus 4 sessions per wk (same volume)	climbing performance	1.	2.	2.
				four different campus board exercises: 1-4-7-10 (Start with both hands on rung 1. Pull through each move and match hands on rung number 10.); Ladder (keep one hand on the start rung and move the other hand up one rung at the time until max reach and Reverse until the hands are matched on rung 1.); 1-2-3 (move ca. 75% of max reach with one hand, then pull through as far as possible with the other hand. Follow with the first hand and match the top.); 10 RM (perform 10 consecutive moves of self-selected length so that the 10th move is near exhaustion.)	Moves completed [#] Campus board performance Completed moves [#] Maximal reach [#] Maximum upper-limb strength Isometric pull-up force [N] Upper-limb power RFD absolute [N/s] RFD ₁₀₀ [N/s]	MS/HYP dyn, Semispecific–2SP +10.2 +61.5 +6.8 +11.3 _a +4.5 –5.6	MS/HYP dyn, Semispecific–4SP +7.8 +69.4 +8.3 +11.7 _a +25.1 +29.3	CG –2 –2.6 –0.4 _a –24.5 –17.6
				1. MS/HYP dyn, semispecific: 4 sets of all exercises twice a wk over 2 days (2SP)				
				2. MS/HYP dyn, semispecific: 4 sets of 2 exercises each session but 4 sessions a wk alternating between exercises 1 and 2, and 3 and 4 (4SP; same volume as group 1)				
				3. CG: climbing training as usual, no campus board allowed				
				All groups maintained regular climbing training				

†f = female; m = male; MS = maximum strength training; HYP = hypertrophy training; END = endurance training; CG = control group; RFD = rate of force development; stat = static exercises; dyn = dynamic exercises; IRCRA = climbing grade according to the international rock climbing research association; ASCF = American sport climbing federation; wk = week; rep. = repetition.

*Significant changes at $p \leq 0.05$, _a = main gains calculated from equal tests on different holds; in studies with multiple posttests, the outcome measures presented refer to the posttest right after the end of the intervention.

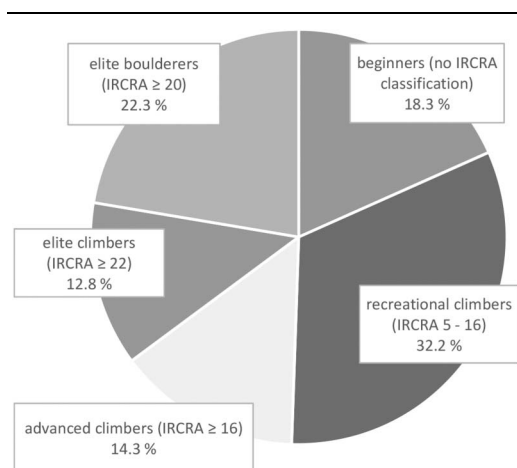


Figure 2. Performance level (climbing or bouldering) of all subjects as a percentage of the total number of subjects [elite boulderers: $n = 61$, elite climbers: $n = 35$, advanced (non) competitive climbers: $n = 35$, recreational climbers: $n = 88$, and beginners: $n = 50$].

Saeterbakken, and Andersen (12) are nonsignificant and only moderate compared with the large effect sizes of the other studies ($d = 0.48$ and $d = 0.73$).

Although López-Rivera and González-Badillo (22) found only nonsignificant improvements in upper-limb strength endurance after 9 weeks of static, semispecific HYP training in elite climbers, López-Rivera and González Badillo (24) found significant improvements for both groups (elite climbers) that also followed a static, semispecific HYP training regimen.

The combined results of all studies incorporated into this review show that different training methods, exercises, and exercise specificity led to improvements in upper-limb strength endurance while a mixture of MS and HYP training including dynamic and static, semispecific exercises led to the largest effect sizes.

Maximum Grip Strength

Of the 5 studies assessing maximum grip strength, 3 showed a significant effect of training for one of the experimental groups (16,17,32). Nonetheless, no significant effect sizes were identified (Figure 5).

Low to moderate effect sizes were found by Jurens (16) ($d = 0.46$) and Kingsley (17) ($d = 0.33$) whose treatment groups followed a MS and HYP training regimen consisting of static, semispecific exercises. A training method similar to that of Kingsley (17) and Jurens (16), performed by Medernach, Kleinöder, and Lötzerich (28), resulted in smaller effects ($d = 0.24$). These findings must, however, be considered against the background of the low PEDro score of the study by Kingsley (17) because of a lack of randomization and specified inclusion criteria. MS training alone, as conducted by Levernier and Laffaye (20), led to an even smaller effect size ($d = 0.2$). The treatment group of the study by Mundry et al. (32) who trained 4 times a week expressed comparatively large effect sizes ($d = 0.44$). However, the other treatment group which trained twice a week according to the same static, semispecific training program however resulted in lower effect sizes ($d = 0.17$).

The combined results of the effect sizes of strength training on maximum grip strength in climbing and bouldering indicate that different training methods, exercises, and exercise specificity lead to an improvement in maximum grip strength. A mixture of MS

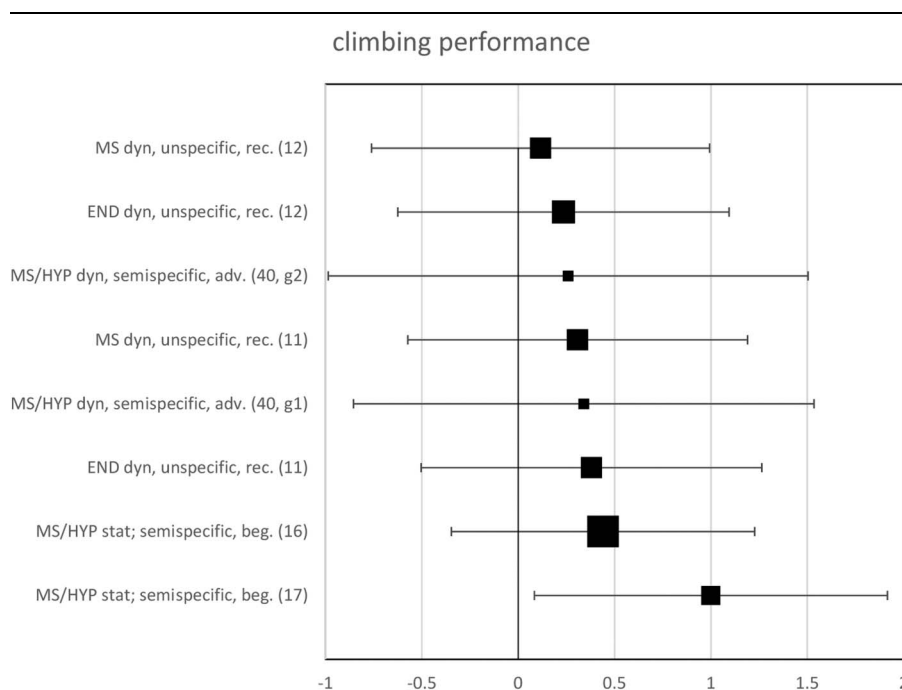


Figure 3. Forest plot with effect sizes and confidence intervals regarding climbing performance of the different treatment groups included into the quantitative analysis. The size of the square represents the respective treatment group's sample size. MS = maximum strength training; HYP = hypertrophy training; END = endurance training; stat = static exercises; dyn = dynamic exercises; beg. = beginners; rec. = recreational climbers; adv. = advanced climbers; g = group; ■ = subjects were climbers.

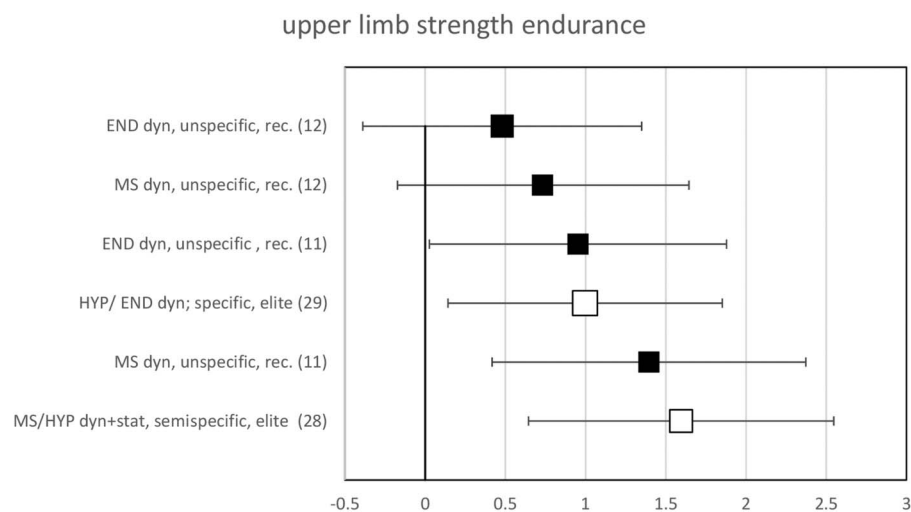


Figure 4. Forest plot with effect sizes and confidence intervals regarding upper-limb strength endurance of the different treatment groups included into the quantitative analysis. The size of the square represents the respective treatment group's sample size. MS = maximum strength training; HYP = hypertrophy training; END = endurance training; stat = static exercises; dyn = dynamic exercises; rec. = recreational climbers; elite = elite athletes; ■ = subjects were climbers; □ = subjects were boulderers.

and HYP training and isolated HYP training including static, semispecific exercises resulted in the largest effect sizes.

López-Rivera and González-Badillo (22) found a non-significant increase in maximum grip strength for both groups (elite climbers) after a HYP training program with static, semi-specific exercises for 8 weeks.

Maximum Upper-limb Strength

Stien et al. (40) conducted the only study assessing maximum upper-limb strength (Figure 6). The results showed positive effect sizes for both treatment groups ($d = 0.45$ and $d = 0.6$). Both groups trained according to a MS and HYP training consisting of

dynamic, semispecific exercises. The effect sizes for both groups however were not different from zero. The treatment group training 4 times per week expresses slightly higher effect sizes compared with the group training only twice per week.

Upper-limb Power

Both isolated MS training (static, semispecific) and a mixture of MS and HYP (dynamic, semispecific) training resulted in positive effect sizes in upper-limb power. However, none of the effect sizes are different from zero (Figure 7). Levernier and Laffaye (20) found larger effect sizes for RFD_{200ms} ($d = 0.96$) compared with RFD_{95%} ($d = 0.39$).

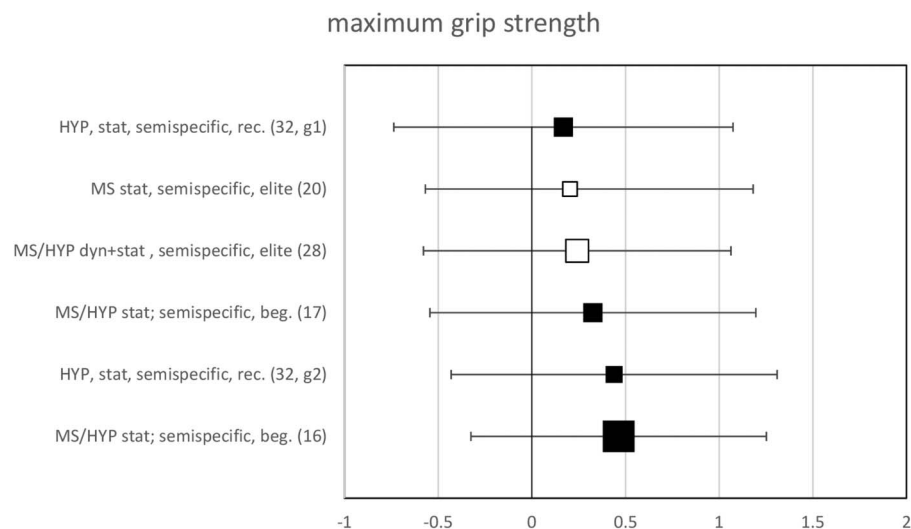


Figure 5. Forest plot with effect sizes and confidence intervals regarding maximum grip strength of the different treatment groups included into the quantitative analysis. The size of the square represents the respective treatment group's sample size. MS = maximum strength training; HYP = hypertrophy training; stat = static exercises; beg. = beginners; rec. = recreational climbers; adv. = advanced climbers; elite = elite athletes; g = group; ■ = subjects were climbers; □ = subjects were boulderers.

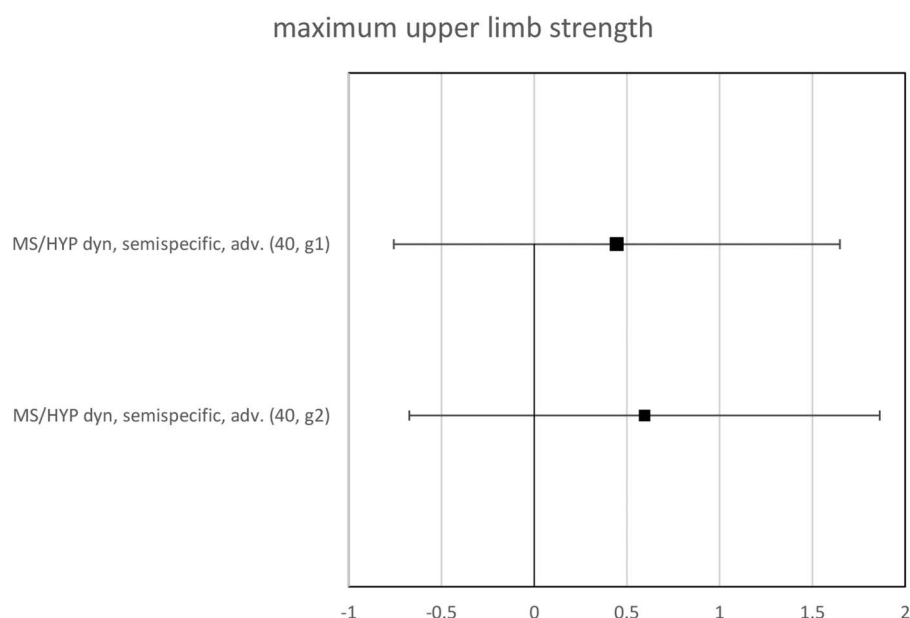


Figure 6. Forest plot with effect sizes and confidence intervals regarding maximum upper-limb strength of the different treatment groups included into the quantitative analysis. The size of the square represents the respective treatment group's sample size. MS = maximum strength training; HYP = hypertrophy training; dyn = dynamic exercises; adv. = advanced climbers; g = group; ■ = subjects were climbers.

Stien et al. (40) however assessed larger effect sizes for $RFD_{absolute}$ ($d = 1.21$ and $d = 0.61$) compared with RFD_{100ms} ($d = 0.75$ and $d = 0.15$) for both treatment groups. The effect sizes of the treatment group training 4 times a week were larger compared with the treatment group training twice a week.

Discussion

With only 12 studies identified, the current state of research on strength training in climbing is rather weak. Because multiple factors such as training method, exercise dynamics and specificity, age, climbing level, experience, and climbing discipline influence the dependent variables, the field of research is large

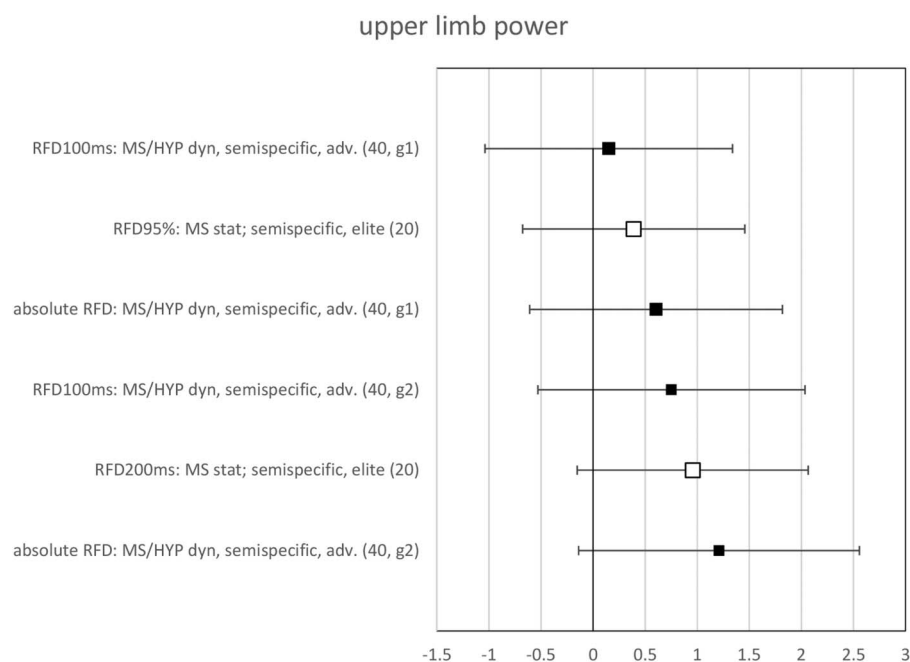


Figure 7. Forest plot with effect sizes and confidence intervals regarding upper-limb power of the different treatment groups included into the quantitative analysis. The size of the square represents the respective treatment group's sample size. MS = maximum strength training; RFD = rate of force development; stat = static exercises; dyn = dynamic exercises; adv. = advanced climbers; elite = elite athletes; g = group; ■ = subjects were climbers; □ = subjects were boulderers.

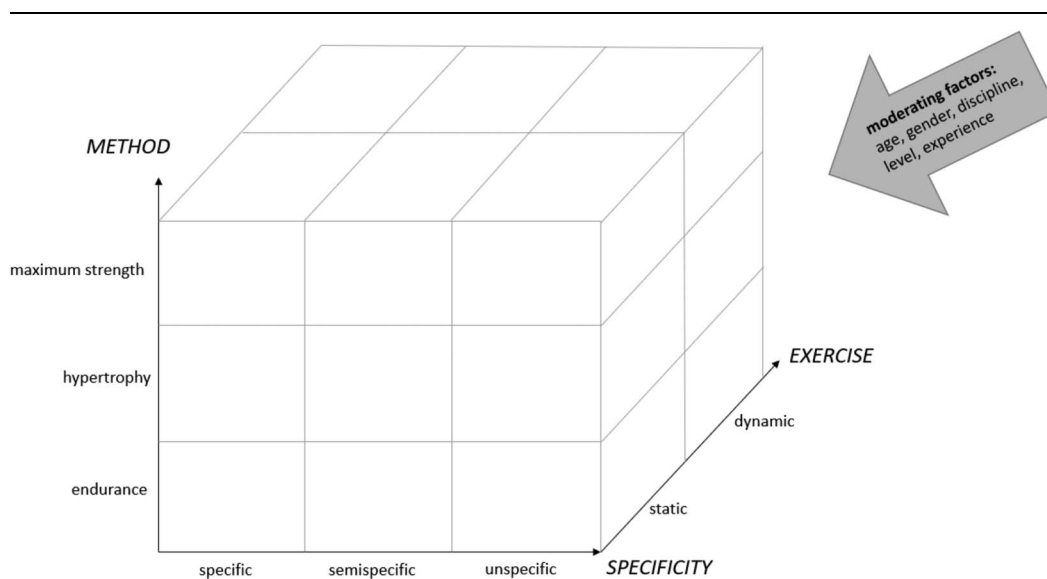


Figure 8. Visualization of the large field of research in climbing.

(Figure 8). Within this high-dimensional space, the number of studies conducted up to now and the sample sizes are very small. Current research is therefore far from allowing to make valid statements. Rather, all results should be treated with caution and cannot be readily generalized.

Climbing and bouldering both consist of a large variety of complex movements influenced by multiple factors such as the incline of the climbing wall, the structure of the hand and foot holds, and the setting of the routes. Owing to its high complexity and the wide range of climbing movements, no direct, valid testing procedure for climbing performance has yet been developed. Although Hermans et al. (11,12) tested climbing performance with an identical artificial climbing route in pretest and posttest that progressively increased in difficulty, the subjects of Jurens (16) and Kingsley (17) had to climb multiple climbing routes with increasing difficulty until exhaustion. Stien et al. (40) assessed the performance of lead climbers by counting the number of completed moves during the best attempt at 2 different boulder problems. This substantial heterogeneity in testing and the high complexity of climbing movements are the reasons why effect sizes regarding climbing performance need to be treated with caution.

The effect sizes calculated for climbing performance ranged from small to large; however, only 1 effect size was significantly different from zero and thus substantial. It is important to note that the largest effect sizes occurred in treatment groups consisting of beginners only. These might have experienced larger gains in climbing performance compared with recreational and advanced climbers because of lower initial technical skill and physical conditions.

In general, the combination of HYP and MS training resulted in higher effect sizes compared with isolated END training. This may be explained by the validity of the tests. Bouldering assessment (40) represents maximum strength factors rather than endurance factors (6,8,41). Regarding climbing performance on one (11,12) or more climbing routes (16,17) with increasing level and intensity, depends more on maximum strength than on other factors.

Although the effect sizes related to unspecific endurance and maximum strength training were negligible or small, the effect sizes for semispecific training ranged from small to large. Semispecific training leading to larger effects in sport performance than unspecific training correlates with the findings of reviews on sprint and overhead ball throw (4,36,43). Although specificity is often neglected to develop general strength and endurance capacities through overload training, sport-specific training is however crucial to transfer strength and endurance abilities to sport performance (4). Brearley and Bishop (4) conclude that in sprint training, a mixed methods approach is most effective to improve general strength through unspecific methods and provides the opportunity to transfer these capacities through specific resistive sport movement training. This position is supported by Rumpf et al. (36) who state that specific sprint training is highly beneficial, whereas nonspecific strength and power training supplement specific training methods and enhance performance. In addition, Van den Tillaar (43) states that specific training with reduced intensity could lead to positive effects in overhead ball throw in elite athletes who are already well conditioned.

This however points toward a problem that is also addressed by Brearley and Bishop (4) who state that transfer strongly depends on the athlete's personal training status because athletes with lower training have more benefits from an unspecific training of muscular capacities compared with highly trained athletes. This points to a differential role of specificity regarding transfer.

Because climbing consists of pure and intermittent isometric contractions and dynamic movements (39), dynamic and static training methods were expected to equally favor climbing performance. However, dynamic training methods led to lower effect sizes in climbing performance compared with static training methods. This seems even more contrainduitive because climbing performance was tested through dynamic movements which would imply a near transfer for dynamic exercises and a farther transfer for static exercises. The fact that most subjects who trained dynamically performed unspecific exercises, and the subjects who trained statically performed semispecific exercises might be a plausible explanation for these findings.

Although only 1 treatment led to effect sizes that were different from zero for climbing performance, 4 effect sizes different from zero were calculated for upper-limb strength endurance. Moreover, the calculated effect sizes were large and considerably larger compared with climbing performance. This can be explained by the fact that the treatments aiming to improve upper-limb strength endurance all included exercises similar to the upper-limb strength endurance tests (specificity effect). This could also explain why semispecific training resulted in larger effect sizes compared with nonspecific exercises.

As stated by Kraemer and Ratamess (18), low training loads effectively lead to an increase in local muscular endurance. Unexpectedly, however, the combination of HYP and MS training led to larger effect sizes than the combination of MS and END training. Furthermore, isolated END training did not lead to significantly larger effect sizes than isolated MS training. The reason for this could be that the tests for upper-limb muscular endurance tended to test maximum strength. In addition, improvements of neural functions (maximum voluntary contraction, rate of force development, and increase of less fatigable Type IIA muscle fibers) could have led to an increase of exercise economy (14).

The fact that some subjects might have had a lower level of upper-limb strength endurance in the pretest because they were boulderers and not climbers, and thus, experienced greater gains can be set aside. According to the findings of Stien et al. (41), there is no significant difference in upper-limb strength endurance between these 2 groups.

As most studies focused on dynamic training while only 1 study combined static and dynamic training (28), no valid statement regarding the quality of the exercises can be derived. In addition, Schöffel et al. (39) state that pure isometric testing of strength endurance in climbing is too unspecific because climbing also involves dynamic and intermittent isometric contractions.

Although the effect sizes calculated for maximum grip strength were significant, they were all small and no effect size was different from zero. According to Baláš et al. (1), the handgrip dynamometer used in all studies to measure maximum grip strength is the most prevalent test. However, studies by Ozimek et al. (33), Giles et al. (9), and Watts et al. (45,46) emphasized that the test conditions of hand dynamometry rarely represent the circumstances of climbing regarding finger and arm position, and the test results should respectively be treated with caution in the context of climbing.

Based on this, Watts et al. (46) recommend a large variability in hand positions during grip strength training leading to a variability in muscle activation. Furthermore, it was stated that a high RFD and contraction velocity are important to maintain contact with handholds (46). This indicates that although climbing requires repetitive isometric contractions of the finger flexors (39), not only static but also dynamic power training of the finger flexors could have positive effects on maximum grip strength and climbing performance.

As already observed for the other dependent variables, a tendency toward a mix of maximal strength and hypertrophy training can be observed. However, this result should be treated with caution as some of the treatment groups of the studies showing the largest effect sizes included beginners and intermediate climbers (16,17,32) while the others included professionals only. The results could therefore rather indicate that elite boulderers with a higher initial level of strength experience

smaller gains compared with climbing beginners with a lower initial level of strength which would support the findings of López-Rivera and Gonzáles-Badillo (23). In addition, regarding the control of intensity, the results of the study by Mundry et al. (32) show that for intermediate climbers adding weight may be superior to reducing edge depth.

Maximum upper-limb strength was only measured by Stien et al. (40). Their results indicate that a mixture of maximum strength and hypertrophy training through dynamic, semispecific exercises resulted in positive effect sizes for maximum upper-limb strength in climbing. However, the effect sizes are not different from zero and thus not substantial.

Nonetheless, a trend toward splitting the training volume over 4 days per week rather than just 2 can be observed. This might be explained by larger fatigue after longer training sessions when training only twice a week (40).

The effect sizes measured for upper-limb power ranged from small to large but did not differ from zero. Several different variables were tested. In addition, the interventions included different training distributions and doses. For this reason, these results cannot be interpreted in a consistent manner. Our hypothesis that benefits for both boulderers and climbers increase with increasing specificity of the training program was only partially confirmed by our findings.

Climbing performance can be increased through both unspecific and semispecific training, whereas semispecific training might result in larger effect sizes. In addition, upper-limb strength endurance in recreational climbers can be increased by unspecific training, whereas semispecific and specific training lead to larger effect sizes in elite boulderers.

Maximum grip strength is only slightly affected by semispecific training; however, climbing novices might profit a little more compared with elite boulderers. On top of that, upper-limb power in elite boulderers and intermediate climbers can also be slightly increased by semispecific strength training.

Nonetheless, much more data on the effects of unspecific, semispecific or specific training on athletes of different levels in bouldering and climbing still need to be gathered to derive valid generalized statements on the most efficient training. We aimed to find out which training methods and parameters are most efficient to increase the performance of climbers and boulderers of different levels.

Although we could not find a solid and final answer to this question, we were able to identify trends. The largest effect sizes in climbing performance and strength endurance have been revealed by studies combining both maximum strength and hypertrophy training. However, the measurements might have been focusing more on maximum strength rather than on endurance factors.

Maximum upper-limb strength, upper-limb power, and maximum grip strength are also positively affected by different training methods. Although isolated MS training has positive effects on the upper-limb power in elite boulderers, the same counts for a mixture of MS and HYP training in intermediate climbers. Maximum upper-limb strength is also positively affected by a mixture of MS and HYP training, but no comparison can be made with other methods.

Both isolated and mixed MS and HYP training also have positive effects on maximum grip strength in boulderers and climbers of different levels while a trend toward the mixture of methods can be observed.

The trend toward a variation of training methods supports the findings of Ratamess et al. (35) who stated that mixed methods

are superior to isolated methods. This can be explained by greater similarity to the target movement through different types of muscle contractions during training.

It is important to keep in mind that we classified the treatments according to the traditional repetition continuum. Regarding this classification, there is no consensus in research (38). A tentative application of a broader stress profile, as suggested by Schoenfeld et al. (38), did not change our findings. If the hypertrophy range for dynamic exercises was for example raised to 8–20 repetitions per set, the second treatment groups of the studies by Hermans et al. (11,12) would be classified as HYP instead of END and the second treatment group of the study by Philippe et al. (34) would be classified as HYP/END instead of END. This would imply that subjects profited more from HYP and MS training combined than from isolated HYP training which is in line with our current results.

Climbing and bouldering both consist of a mixture of dynamic and static movements which is why both static and dynamic training are likely to improve climbing performance. Our results show that a combination of dynamic and static exercises leads to positive effect sizes in maximum grip strength and to exceptionally large effect sizes in upper-limb strength endurance which confirms this assumption.

Furthermore, our results show that both dynamic and static training methods lead to an increase in climbing performance. Although static training seems to lead to larger effect sizes than dynamic training, this assumption needs to be treated with caution because the subjects training statically were beginners and could thus have experienced a larger training effect compared with recreational climbers.

It can also be concluded that dynamic exercises lead to positive effects in upper-limb strength endurance while both dynamic and static exercises lead to positive effects in upper-limb power. In addition, dynamic exercises lead to an increase in maximum upper-limb strength, and static exercises lead to an increase in maximum grip strength. Whether static or dynamic training, or a mixture of both, favors boulderers or climbers of certain levels more than the other cannot be decided based on our results.

Practical Applications

There are several practical conclusions that can be drawn from the findings of this review and that can help coaches and athletes to optimize their training. Owing to a fragmentary and rather sparse state of research, our findings can however not be generalized without consideration. Our review shows that training recommendations of all sorts should be treated with caution because no generalized findings have yet been reported by exercise science. It can however be said that coaches and athletes should focus on mixing hypertrophy and maximum strength training to increase upper-limb strength endurance, maximum grip strength, and climbing performance. Furthermore, it is very likely that a mixture of both dynamic and static training will lead to the largest training effects. In addition, athletes should focus on semispecific training methods. However, these findings do not contradict unspecific training especially for untrained individuals which leads to smaller effects but is likely to benefit the overall strength and climbing performance.

Acknowledgments

There were no professional relationships with companies or manufacturers that will benefit from the results of this review. There was no specific financial support for the preparation of this review. The results of the present review do not represent an endorsement of the product by the authors or the NSCA. The authors have no potential conflicts of interest that are directly related to this review.

References

- Baláš J, Panáčková M, Kodejška J, Cochrane JD, Martin JA. The role of arm position during finger flexor strength measurement in sport climbers. *Int J Perform Anal* 14: 345–354, 2014.
- Baláš J, Pecha O, Martin AJ, Cochrane D. Hand–arm strength and endurance as predictors of climbing performance. *Eur J Sport Sci* 12: 16–25, 2012.
- Borenstein M, Hedges LV, Higgins JPT, Rothstein HR. When does it make sense to perform a meta-analysis? In: *Introduction to Meta-Analysis*. Padstow Cornwall: Wiley and Sons Ltd, 2009. pp. 357–364.
- Brearley S, Bishop C. Transfer of training: How specific should we be? *Strength Cond J* 41: 97–109, 2019.
- Draper N, Giles D, Schöffl V, et al. Comparative grading scales, statistical analyses, climber descriptors and ability grouping: International Rock Climbing Research Association position statement. *Sports Technol* 8: 88–94, 2015.
- Fanchini M, Violette F, Impellizzeri FM, Maffiuletti NA. Differences in climbing-specific strength between boulder and lead rock climbers. *J Strength Cond Res* 27: 310–314, 2013.
- Fritz CO, Morris PE, Richler JJ. Effect size estimates: Current use, calculations, and interpretation. *J Exp Psychol* 141: 2–18, 2012.
- Fryer S, Stone KJ, Sveen J, et al. Differences in forearm strength, endurance, and hemodynamic kinetics between male boulderers and lead rock climbers. *Eur J Sport Sci* 17: 1177–1183, 2017.
- Giles LV, Rhodes EC, Taunton JE. The physiology of rock climbing. *Sports Med* 36: 529–545, 2006.
- Guolo A, Varin C. Random-effects meta-analysis: The number of studies matters. *Stat Methods Med Res* 26: 1500–1518, 2017.
- Hermans E, Andersen V, Saeterbakken AH. The effects of high resistance–few repetitions and low resistance–high repetitions resistance training on climbing performance. *Eur J Sport Sci* 17: 378–385, 2017.
- Hermans E, Saeterbakken AH, Andersen V. Effect of maximal- and local muscular endurance strength training on climbing performance and climbing-specific strength in recreational climbers: A randomized controlled trial. In: *Proceedings of the 3rd Rock Climbing Research Congress*. Telluride, Colorado, School of Health & Human Performance. Draper N, Drum S, Joubert L, and Watts P, eds. Northern Michigan University, 2016. pp. 38–39.
- Hörst EJ. *Training for Climbing: The Definitive Guide to Improving Your Performance*. Guilford, CT, Helena, MT: Falcon, 2008.
- Hughes D, Ellefsen S, Baar. Adaptations to endurance and strength training. *Cold Spring Harb Perspect Med* 8: a029769, 2018.
- International Olympic Committee (IOC). *International Federations. In: IOC Annual Report 2016: Credibility, Sustainability and Youth*. Lausanne, Switzerland: International Olympic Committee, 2016. pp. 20–23.
- Jurrens JD. *The Effects of Hang Board Exercise on Grip Strength and Climbing Performance in College Age Male Indoor Rock Climbers [Master's thesis]*. La Crosse: University of Wisconsin, 1997. pp. 28–47.
- Kingsley AM. *The Effects of Hangboard Exercise on Climbing Performance and Grip Strength in College Female Indoor Rock Climbers [Master's thesis]*. La Crosse: University of Wisconsin, 1997. pp. 21–37.
- Kraemer WJ, Ratamess NA. Fundamentals of resistance training: Progression and exercise prescription. *Med Sci Sports Exerc* 36: 674–688, 2004.
- Laffaye G, Levernier G, Collin JM. Determinant factors in climbing ability: Influence of strength, anthropometry, and neuromuscular fatigue. *Scand J Med Sci Sports* 26: 1151–1159, 2016.
- Levernier G, Laffaye G. Four weeks of finger grip training increases the rate of force development and the maximal force in elite and top world-ranking climbers. *J Strength Cond Res* 33: 2471–2480, 2019.
- Lin J, Chen T. Diversity of strength training methods: A theoretical approach. *Strength Cond J* 34: 42–49, 2012.
- López-Rivera E, González-Badillo JJ. The effects of two maximum grip strength training methods using the same effort duration and different edge depth on grip endurance in elite climbers. *Sports Technol* 5: 100–110, 2012.

23. López-Rivera E, González-Badillo JJ, Draper N, Romero NE, Fuss T. The effects of a weighted dead-hang training program on grip strength and endurance in expert climbers with different levels of strength. In: *Proceedings of the 4th Rock Climbing Research Congress Chamonix*. Draper N, Romero NE, Fuss T, et al., eds. Chamonix, France, Ecole Nationale de Ski et d'Alpinisme, 2018. p. 15.
24. López-Rivera E, González-Badillo JJ. Comparison of the effects of three hangboard strength and endurance training programs on grip endurance in sport climbers. *J Hum Kinet* 66: 183–195, 2019.
25. Lum D, Barbosa TM. Brief review: Effects of isometric strength training on strength and dynamic performance. *Int J Sports Med* 40: 363–375, 2019.
26. Macdonald JH, Callender N. Athletic profile of highly accomplished boulderers. *Wilderness Environ Med* 22: 140–143, 2011.
27. Maher C, Sherrington C, Herbert R, Moseley A, Elkins M. Reliability of the PEDro scale for rating quality of randomized controlled trials. *Phys Ther* 83: 713–721, 2003.
28. Medernach JPJ, Kleinöder H, Lötzerich HHH. Fingerboard in competitive bouldering: Training effects on grip strength and endurance. *J Strength Cond Res* 29: 2286–2295, 2015.
29. Medernach JP, Kleinöder H, Lötzerich HHH. Effect of interval bouldering on hanging and climbing time to exhaustion. *Sports Technol* 8: 76–82, 2015.
30. Mermier CM, Janot JM, Parker DL, Swan JG. Physiological and anthropometric determinants of sport climbing performance. *Br J Sports Med* 34: 359–365, 2000.
31. Michailov ML. Workload characteristic, performance limiting factors and methods for strength and endurance training in rock climbing. *Med Sport* 11: 97–106, 2014.
32. Mundry S, Steinmetz G, Atkinson EJ, Schilling AF, Schöffl VR, Saul D. Hangboard training in advanced climbers: A randomized controlled trial. *Sci Rep* 11: 1–8, 2021.
33. Ozimek M, Staszkiwicz R, Rokowski R, Stanula A. Analysis of tests evaluating sport climbers' strength and isometric endurance. *J Hum Kinet* 53: 249–260, 2016.
34. Philippe M, Filzwieser I, Leichtfried V, et al. The effects of 8 weeks of two different training methods on on-sight lead climbing performance. *J Sports Med Phys Fitness* 59: 561–568, 2019.
35. Ratamess NA, Alvar BA, Evetovich TK, et al. Progression models in resistance training for healthy adults. *Med Sci Sports Exerc* 41: 687–708, 2009.
36. Rumpf M, Lockie RG, Cronin JB, Jalilvand F. Effect of different sprint training methods on sprint performance over various distances: A brief review. *J Strength Cond Res* 30: 1767–1785, 2016.
37. Saul D, Steinmetz G, Lehmann W, Schilling A. Determinants for success in climbing: A systematic review. *J Exerc Sci Fitness* 17: 91–100, 2019.
38. Schoenfeld BJ, Grgic J, Van Every DW, Plotkin DL. Loading recommendations for muscle strength, hypertrophy, and local endurance: A Re-examination of the repetition continuum. *Sports* 9: 1–25, 2021.
39. Schöffl V, Möckel F, Köstermeyer G, Roloff I, Küpper T. Development of a performance diagnosis of the anaerobic strength endurance of the forearm flexor muscles in sport climbing. *Int J Sports Med* 27: 205–211, 2005.
40. Stien N, Pedersen H, Vereide VA, et al. Effects of two vs. four weekly campus board training sessions on bouldering performance and climbing-specific tests in advanced and elite climbers. *J Sports Sci Med* 20: 438–447, 2020.
41. Stien N, Saeterbakken AH, Hermans E, Vereide VA, Olsen E, Andersen V. Comparison of climbing-specific strength and endurance between lead and boulder climbers. *PLoS One* 14: 1–13, 2019.
42. Valentine JC, Pigott TD, Rothstein HR. How many studies do you need? A primer on statistical power for meta-analysis. *J Educ Behav Stat* 35: 215–247, 2010.
43. Van Den Tillaar R. Effect of different training programs on the velocity of overarm throwing: A brief review. *J Strength Cond Res* 18: 388–396, 2004.
44. Vereide V, Kalland J, Solbraa AK, Andersen V, Saeterbakken AH. Correlation between relative peak-, isometric force and RFD and climbing performance. In: *Proceedings of the 3rd Rock Climbing Research Congress*. Telluride, Colorado, School of Health & Human Performance. Draper N, Drum S, Joubert L, and Watts P, eds. Northern Michigan University, 2016. pp. 12–13.
45. Watts PB. Physiology of difficult rock climbing. *Eur J Appl Phys* 91: 361–372, 2004.
46. Watts PB, Jensen RL, Gannon E, Kobeinia R, Marnard J, Sansom J. Forearm EMG during rock climbing differs from EMG during handgrip dynamometry. *Int J Exerc Sci* 1: 4–13, 2008.
47. Wong EKL, Ng GYF. Strength profiles of shoulder rotators in healthy sport climbers and non climbers. *J Athl Train* 44: 527–530, 2009.